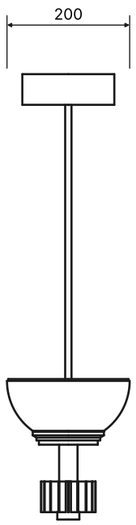
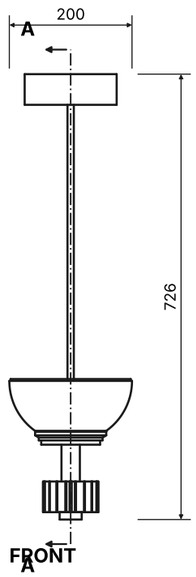
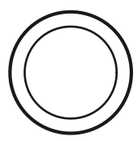
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

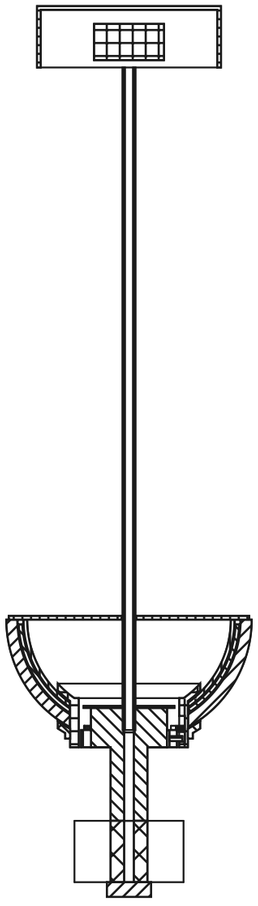
General arrangement

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. Section A-A (sheet 1): the product cut at x 0.0 and seen from the right; cut faces hatched, a different angle for each part.



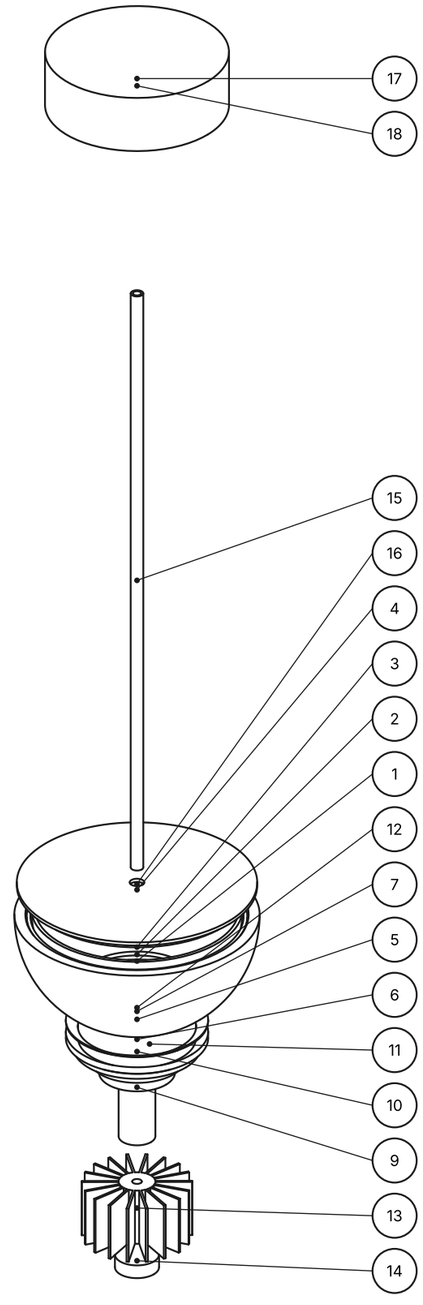
Section A-A

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5, parts drawn apart. Balloons are the parts list's numbers (last sheet).



Exploded view

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

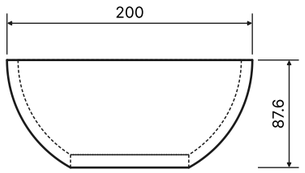
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

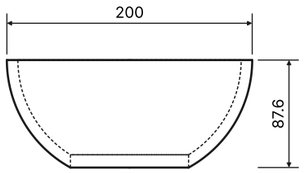
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

1. shell: formed, acrylic-formed; 446 cm3 each; 1 for the set; 1:5

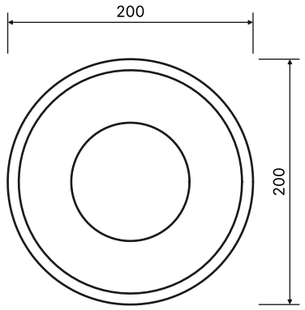
the outside: 10 mm tinted cast sheet drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds (9 at the crown, thinner toward the rim: the former confirms); its outer face never touches the mould, so it keeps the cast sheet's gloss



FRONT



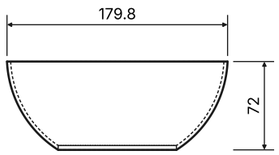
FROM THE RIGHT



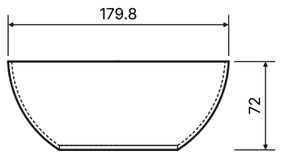
FROM ABOVE

2. liner: formed, acrylic-opal-formed; 117 cm3 each; 1 for the set; 1:5

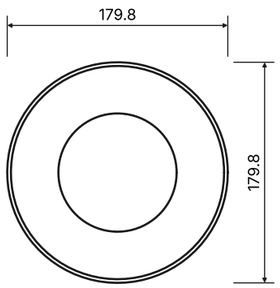
the diffuser: 3 mm volume-diffusing (opal) cast acrylic, light-diffusing grade, about 70 % transmission drape-formed from cast sheet over an aluminium mould, trimmed and polished; the prototypes are formed on the production moulds. Opal scatters through its volume, so it diffuses wherever it touches the tinted layers (a bead-blasted clear liner goes clear where it is bonded or wetted)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

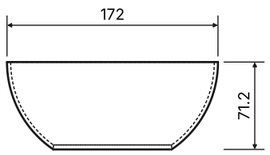
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

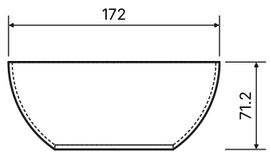
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

3. skin: formed, acrylic-formed; 111 cm3 each; 1 for the set; 1:5

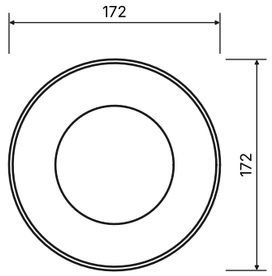
the inside: 3 mm tinted cast sheet formed into a female mould, so its inner (seen) face keeps its gloss and the opal is seen only through tint



FRONT



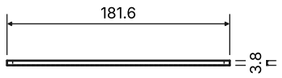
FROM THE RIGHT



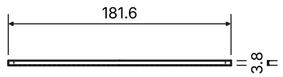
FROM ABOVE

4. rim: lathe, cast-acrylic; 10 cm3 each; 1 for the set; 1:5

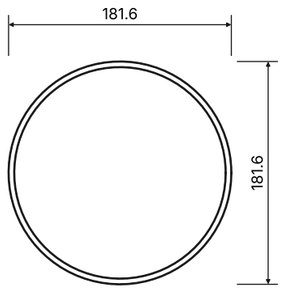
closes the three layers at the mouth, bonded and polished with them; a press-in ePTFE vent in its face keeps the gaps between the layers at room pressure and dry



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

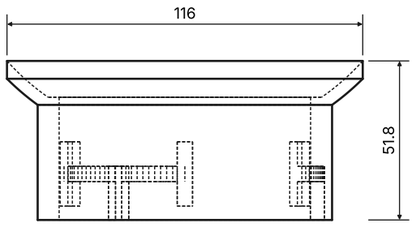
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

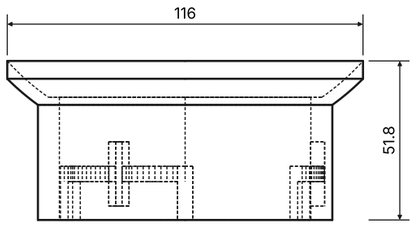
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

5. collar: lathe, aluminium; 94 cm3 each; 1 for the set; 1:2

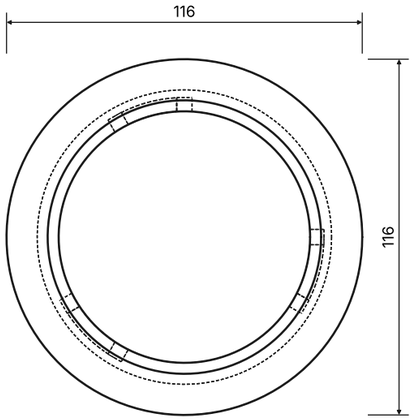
aluminium 6061, black anodised, satin: a tube through the crown with a spherical flange under the skin; three blind L-slots with T-ended notches (the quarter-turn bayonet, 30 deg) work either way up; M96 × 1.5 outside for the clamp ring. It carries the cover mechanically: the bond is a seal only



FRONT



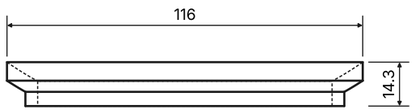
FROM THE RIGHT



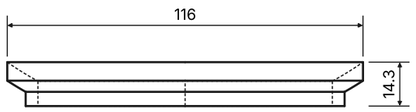
FROM ABOVE

6. clamp: lathe, aluminium; 24 cm3 each; 1 for the set; 1:2

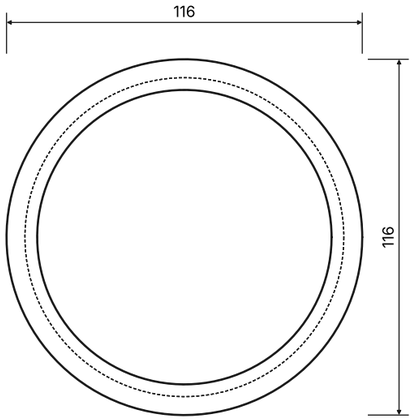
aluminium, black anodised: screwed onto the collar over the crown, its spherical face on a silicone ring, to a torque the shade maker sets (the acrylic is clamped between two silicone rings, never metal on acrylic); a drop of thread-lock



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

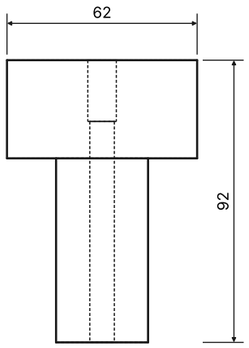
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

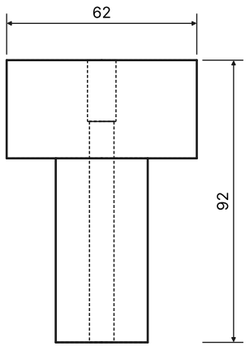
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

9. hub: lathe, aluminium; 134 cm3 each; 1 for the set; 1:2

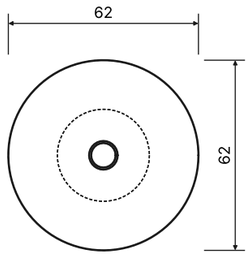
aluminium 6061, black anodised: the core, turned from bar; the LED ring on its face (inside the collar, facing up into the bowl), the PEEK ring round it, a 60 neck down to the fins, threaded for the rod (1/8 IPS) at its top



FRONT



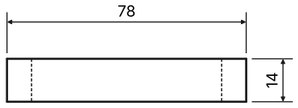
FROM THE RIGHT



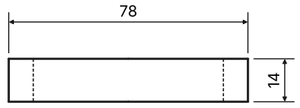
FROM ABOVE

10. ring: lathe, peek; 25 cm3 each; 1 for the set; 1:2

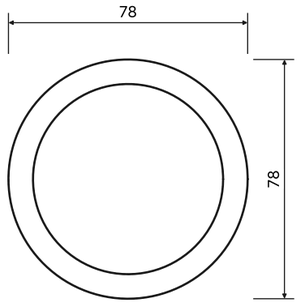
PEEK, unfilled: the thermal break, as North's; it carries the pins and the detent



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

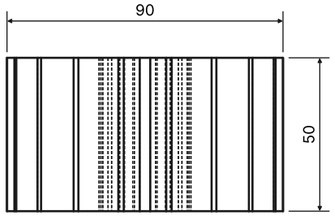
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

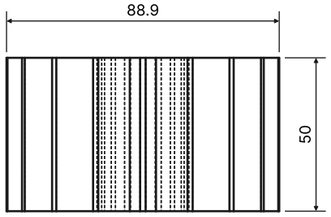
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

13. fins: extrusion, aluminium; 81 cm3 each; 1 for the set; 1:2

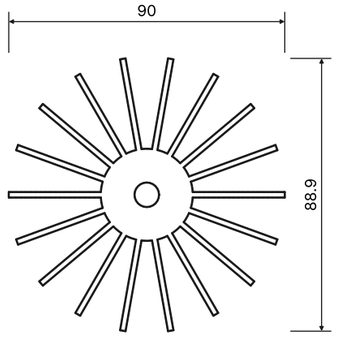
the heat sink below the bowl, in room air: 50 cut from a radial extrusion 90 across, 18 fins, black anodised; 60 below the crown on the neck, so its warm air has mixed before it reaches the acrylic



FRONT



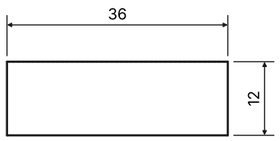
FROM THE RIGHT



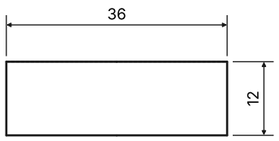
FROM ABOVE

14. finial: lathe, cast-acrylic; 12 cm3 each; 1 for the set; 1:1

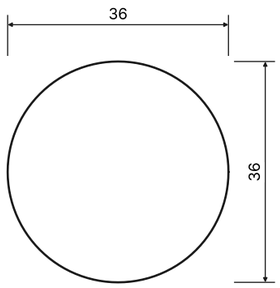
black cast acrylic, polished: closes the fins' core at the bottom (a 1930s bowl's finial)



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

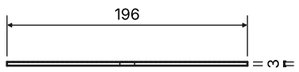
mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

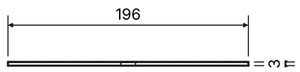
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

16. lid: mill, cast-acrylic; 90 cm3 each; 1 for the set; 1:5

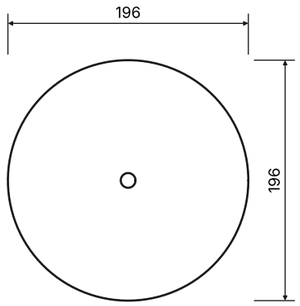
3 mm clear cast acrylic, bead-blasted both faces (or a high-transmission opal): the dust lid, lying on the rim on three small clear silicone bumpers; it lifts up the rod to be wiped. Its light is the ceiling wash



FRONT



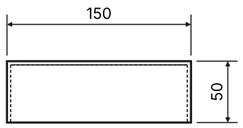
FROM THE RIGHT



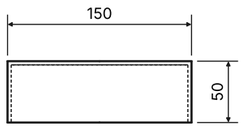
FROM ABOVE

17. canopy: lathe, cast-acrylic; 118 cm3 each; 1 for the set; 1:5

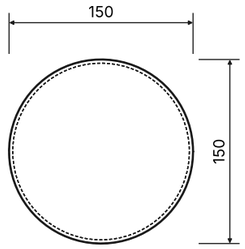
150 × 50, black cast acrylic, polished, as North's: over the driver and a steel junction-box strap; the rod hangs from the strap through a listed swivel (a hickey) so it hangs plumb



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

mm | issue 2026-10-22 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	shell	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	446 cm3, 210 × 210 × 98	1,534.41
2	liner	formed	1	Cast acrylic sheet, volume-diffusing (opal), formed over an aluminium mould	117 cm3, 190 × 190 × 82	1,212.73
3	skin	formed	1	Cast acrylic sheet, tinted through, drape-formed over an aluminium mould	111 cm3, 182 × 182 × 81	1,212.30
4	rim	lathe	1	Cast acrylic tube, tinted through	192 × 192 × 14	65.26
5	collar	lathe	1	Aluminium 6061	126 × 126 × 62	69.55
6	clamp	lathe	1	Aluminium 6061	126 × 126 × 24	54.65
7	gaskets	buy	1	Two moulded silicone rings (Shore 40, black, 1 mm, spherical): one between the collar's flange and the skin inside, one between the clamp ring and the shell outside		3.00
8	vent	buy	1	Gore or alike A press-in ePTFE pressure-equalising vent (about 4 across), in the rim ring: it lets the air gaps between the layers breathe and keeps out dust and liquid		1.50
9	hub	lathe	1	Aluminium 6061	72 × 72 × 102	58.22
10	ring	lathe	1	PEEK rod, unfilled	88 × 88 × 24	110.05
11	pins	buy	1	Three stainless dowel pins 4 × 8 (ISO 8734), pressed into the hub: the bayonet's pins		1.50
12	led	buy	1	to make (an LED board house), with Samsung LM301H-class 3030 LEDs Hemisphere South's light: a ring MCPCB 74 × 14 × 1.6 (aluminium core), 2700 K CRI 90 mid-power LEDs in strings matched to the driver's window: about 2,200 lm at about 14.7		14.00
13	fins	extrusion	1	Aluminium 6061	100 × 99 × 60	19.84
14	finial	lathe	1	Cast acrylic tube, tinted through	46 × 46 × 22	46.86
15	rod	buy	1	any lighting supplier 1/8 IPS threaded steel lamp pipe, 10.3 across, black, cut to the drop, with a listed swivel (hickey) and a locknut at the canopy; the wires (18 AWG, listed fixture wire, Class 2 DC) inside		6.00
16	lid	mill	1	Cast acrylic tube, tinted through	206 × 206 × 13	67.07
17	canopy	lathe	1	Cast acrylic tube, tinted through	160 × 160 × 60	106.44
18	driver	buy	1	Mean Well (to confirm against the board) A Class 2 constant-current, phase-cut dimmable driver, 120 V in, for the ring board: PCD-25 class for the 2,200 lm level (about 15 W), PCD-16 class for the 1,150 lm level (about 8 W); the board's strings		24.00
19	plunger	buy	1	Kipp or Misumi A stainless ball spring plunger M5, light force (about 10 N), threaded into the hub: the bayonet's detent, clicking into a dimple in the collar		4.50
20	listing	buy	1	UL or Intertek (ETL) Listing the pendant to UL 1598 (and the table lamp's battery and charger to UL 153 / IEC 62368): a one-off cost a design, spread over the first run		[PRICE]

Parts list

Hemisphere South: Uplighter bowl on a rod in cast acrylic, its shades swapping by hand

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Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. The same shades as North, turned over: a bowl, opening up, on a slim rod from the ceiling. Any shade fits either fixture: the collar's slots end in a T, so the pins rest up in the notch when the cover hangs (North) and down in it when the cover sits (South); its own weight holds it either way.
2. The light: a ring of mid-power LEDs on the hub's face, round the rod, facing up into the bowl. About 57 % of it leaves through the mouth and the frosted lid as a soft wash on the ceiling; the rest falls on the bowl's wall, which glows in its tint. Bounced light needs about twice the lumens to feel as bright, so South gives twice North's.
3. The heat: the fins hang below the bowl in room air on a 60 neck; the PEEK ring keeps the hub off the collar. See hemisphere_thermal.py and the test plan (README, round 35).
4. The dust lid: 3 mm frosted acrylic on the rim, lifted up the rod to wipe.
5. Hang with the rim 450 to 600 below a ceiling of 2.6 to 3.2 m, so the wash spreads; not over a table (its light is on the ceiling).

Notes

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